

Numerical modeling of fiber orientation in additively manufactured composites

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Abstract. Additive manufacturing has undergone a significant transformation, evolving from a mere prototyping technique to a reliable and proven manufacturing technology that can produce products of varying sizes and materials. The incorporation of fibers in additive manufacturing processes has the potential to improve a range of material properties, including mechanical, thermal, and electrical properties. However, this improvement is largely dependent on the orientation of the fibers within the material, with the properties being enhanced primarily in the direction of fiber orientation. As a result, accurately predicting and controlling the fiber orientation during the extrusion or deposition process is critical. Various methods are available to control fiber orientation, such as manipulating the nozzle shape, extrusion and nozzle speed, the gap between the nozzle and substrate, as well as fiber features like aspect ratio and volume fraction. At the same time, the presence and orientation of fibers can significantly impact the flow pattern and extrusion pressure conditions, ultimately affecting the formation of printed strands in a manner distinct from those without fibers. For that reason, our study utilizes computational fluid dynamics to anticipate and comprehend the printing conditions that would result in favorable fiber orientations and strand shapes, incl. corner printing. Our findings may be utilized to determine optimal toolpaths for 3D printing composites, as well as printing conditions that will facilitate the achievement of the desired fiber orientation within individual strands.

1. Introduction

Fused-deposition modeling (FDM), also known as Material Extrusion Additive Manufacturing (MEX-AM), has gained significant traction as a highly efficient technology for producing micro-components and large-scale structures. Its adaptability has led to widespread adoption in diverse sectors such as energy, construction, medicine, aerospace, bio-composites, and consumer products (1). Furthermore, significant attention is presently dedicated to exploring the potential of additive manufacturing in the design of wind turbine blades for large rotors, with a specific emphasis on blade tips. Additionally, recycling fibers extracted from wind turbine blades holds promise as innovative filaments suitable for additive manufacturing (2). As FDM has recently transitioned from a prototyping technique to industrial-scale manufacturing, it must meet stringent quality requirements for various applications. Consequently, in certain instances, relying solely on a single component material, such as polymer, may prove insufficient,



necessitating the addition of fibers to reinforce the material and enhance its thermal, electrical, and mechanical properties (3). Given the considerable freedom offered by 3D printing in manufacturing parts, the specific printing method employed significantly influences the orientation of the fibers within the material. As fiber orientation plays a vital role in improving mechanical and other thermo-physical properties, it becomes a crucial consideration in the discussion (4).

Numerical simulations using computational fluid dynamics have been commonly employed to predict the flow of molten polymer and understand the formation of 3D printed strands (5; 6; 7). However, there has been a limited amount of research focusing on describing the behavior of fibers during 3D printing, particularly regarding their orientation. Previous 2D models have identified the existence of core/skin regions, but they were unable to specify the spatial distribution of these zones (8; 9). Only recently, the first 3D numerical simulation of the additive manufacturing process known as compression molding was published, demonstrating the presence of skin/core regions and their spatial distribution under certain printing conditions (10; 11; 12). These simulations were further complemented by parametric studies conducted by different authors, who explored the influence of various printing conditions on both strand morphology and fiber orientation (13; 14; 15). Additionally, understanding fiber orientation during multi-layer 3D printing has been crucial, as previously printed strands, especially in large-area additive manufacturing, tend to experience deformation that can cause fiber reorientation (16). Experimental studies have also contributed to this body of knowledge, particularly highlighting the significance of strand width as a critical factor correlated with fiber orientation. It has been observed that wider strands exhibit lower fiber orientation in the printing direction (17). Furthermore, Yan et al. have explored fiber orientation in common 3D printed shapes, such as corners, T-junctions, and sinusoidal waves (18).

The present study aims to utilize numerical simulations to predict fiber orientation in both single-layer and multi-layer printing scenarios, with a specific focus on corner prints with a 90° angle. The manuscript begins by describing the numerical model implemented in OpenFOAM. Subsequently, the results are discussed in relation to single-layer printing, multi-layer printing, and corner prints. The manuscript concludes with a summary of our findings.

2. Numerical model

The simulation model employed in this study involved two-phase fluid simulations, with one phase representing the extrusion/deposition material (such as a thermoplastic or thermoset polymer, concrete, etc.) and the other phase representing air. The open-source software OpenFOAM was utilized, and specifically, the two-phase solver called overInterDyMFOam, based on the volume-of-fluid method with overset methodology enabled, was chosen as the initial solver. Additionally, since the material was reinforced with fibers, a macroscopic tensor approach was employed to simulate their orientation. The governing equations for mass and momentum, along with the transport equations for the phase-scalar α and the orientation tensor, are provided below:

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = \rho \mathbf{g} + \nabla \cdot \boldsymbol{\sigma} + \mathbf{f}_{\sigma_i} \quad (2)$$

where ρ is the density of material matrix, $\mathbf{u}$ is the velocity vector, $\mathbf{g}$ is the gravity vector, $\boldsymbol{\sigma}$ is the stress, $\mathbf{f}_{\sigma_i}$ is the surface tension modeled as continuum surface force and ∇ is the gradient operator.

The stress tensor, for the matrix material reinforced by fibers, can be written as (13; 16):

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\mu_{app}(\mathbf{D} + N_p\mathbf{A}_{(4)}\mathbf{D}) \quad (3)$$

where p is the pressure, μ_{app} is the apparent dynamic viscosity, $\mathbf{D} = ((\nabla \mathbf{u}) + (\nabla \mathbf{u})^T)/2$ is deformation rate tensor and $A_{(4)}$ is fourth-rank fiber orientation tensor. As the problem is two-way coupled, the influence of fiber on flow dynamics through stress term in the momentum equation is given through the term $N_p \mathbf{A}_{(4)} \mathbf{D}$, where N_p is the level of anisotropy dependent on fiber volume fraction and aspect ratio.

The volume-of-fluid approach considers that a phase-scalar, α , should be transported to delineate between two phases. The transport equation of α can be written as:

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{u}) + \nabla \cdot [\alpha \mathbf{u}_c (1 - \alpha)] = 0 \quad (4)$$

where α is the phase fraction of reinforced matrix material, $1 - \alpha$ is the fraction of surrounding air and $\mathbf{u}_c$ is the interface compression velocity which controls the diffusive interface (19).

In this study, the material matrix was considered to exhibit non-Newtonian behavior, and specifically, the Bingham model was utilized to effectively control the deformation of the bottom strand. The Bingham model is commonly employed in large-scale additive manufacturing applications, particularly for modeling the flow of materials like concrete (20; 14; 21; 22), thermoset (23) or other substances that experience changes in yield stress over time. Hence, while this study and methodology have the potential for broad applicability across different matrix materials, they are specifically designed for thermoset materials. The mathematical representation of the Bingham model is as follows:

$$\mu_{app} = k + \frac{\tau}{\dot{\gamma}} \quad \text{for} \quad \dot{\gamma} \geq \dot{\gamma}_c \quad (5)$$

where k is the plastic viscosity, τ is the yield stress, $\dot{\gamma}$ is the shear rate and $\dot{\gamma}_c$ is the critical shear rate. In this study, $\dot{\gamma}_c$ was set to 0.01, similarly as in (20). It has to be highlighted here that fiber presence and orientation did not influence yield stress value as in the novel model presented in (24), and the yield stress value was assumed to be constant. The presence of fibers did not directly impact the viscosity term μ_{app} as in (25). Instead, their influence was incorporated as an additional term in the stress formulation, as shown in Equation 3.

The equation used to determine fiber orientation is expressed as the following tensor transport equation:

$$\frac{\partial \mathbf{A}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{A} = -(\mathbf{W} \cdot \mathbf{A} - \mathbf{A} \cdot \mathbf{W}) + \lambda_F (\mathbf{D} \cdot \mathbf{A} + \mathbf{A} \cdot \mathbf{D} - 2\mathbf{D} : \mathbf{A}_{(4)}) + 2D_r (\mathbf{I} - 3\mathbf{A}) \quad (6)$$

where $\mathbf{W} = ((\nabla \mathbf{u}) - (\nabla \mathbf{u})^T)/2$ is the vorticity tensor, D_r is the rotary diffusion coefficient accounting for fiber-fiber interaction, $\mathbf{I}$ is the unit tensor and λ_F is the shape factor of fibers. Further information regarding the fiber modeling approach can be found in (11; 13; 16). These references provide comprehensive details on the numerical simulations, including the specific initial and boundary conditions utilized in the study.

3. Results and discussion

In this section, it will be investigated how fiber orientates during different situations encountered during 3D printing. For example, by changing different processing parameters, the shape of the strand cross-section will change, and this will influence fiber orientation. On the other hand, the majority of printed structures are composed of multiple layers, where layers can be built before the first deposited layer is completely solidified. This will result in slight deformation of previously deposited layers, and hence fiber reorientation. Moreover, it will be discussed how fiber orientation during corner printing can be used in the optimization of the toolpath wrt. to the fiber orientation. The effect of viscosity and fiber geometry are studies elsewhere (13; 16).

3.1. Effect of printing conditions on fiber orientation

As explained in a previous study (13), the fiber orientation in 3D printing is highly influenced by various printing conditions, particularly the ratio of extrusion/nozzle speed. When the nozzle moves slowly or the material is extruded rapidly, the resulting strand shape tends to become more rectangular, which promotes lateral flow of the material. This lateral flow occurs as a squeezing effect between the substrate and the nozzle exit, leading to deformation gradients along the height of the strand. These gradients cause the reorientation of fibers and enhance their alignment in the lateral direction.

It is important to note that all printing conditions, including gap height, nozzle/extrusion speed, diameters, matrix rheology, fiber aspect ratio, and volume fraction, impact the resulting fiber orientation and overall material strength. A numerical model is used to quantify the extent of these effects and identify the most influential parameters in this regard.

To illustrate the impact of printing conditions on fiber orientation, we considered an example of different gap heights. Although previous research (13) suggests that gap height affects the distribution of fiber orientation, with fibers generally being similarly oriented in the cross-section on average, it serves as a useful example to demonstrate the noticeable differences in strand appearance. In our study, we employed the Bingham model, which is applicable to a range of materials exhibiting viscoplastic behavior with a variable yield stress value threshold that may change over time (e.g., during curing, hardening, solidification, etc.).

Initially, it is important to recognize that fibers primarily align in the extrusion direction (z -direction) within the nozzle Fig. 1. Therefore, if no deformation occurs during the placement of the strand onto the substrate, these z -oriented fibers would be aligned in the x -direction. However, due to the presence of deformation to some extent, whether intentional or unintentional, the fiber orientation observed within the nozzle is altered, resulting in a new distribution of fiber orientations in the strand. This emphasizes the significant role played by printing conditions, as the deviation from a perfectly circular strand is influenced by these conditions as well as material and gravity forces.

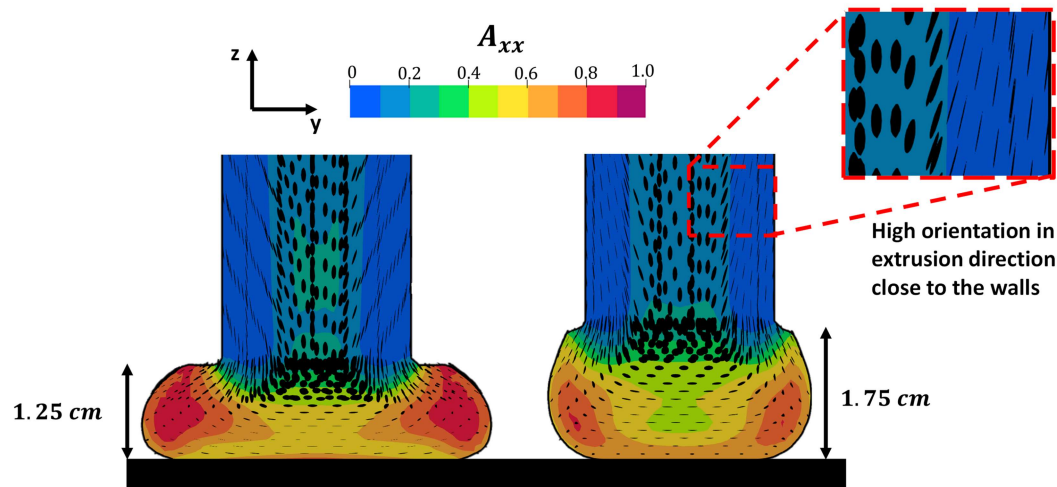


Figure 1: Fiber orientation at the end of the nozzle and during strand deposition.

In Fig. 1, it is evident that the lateral flow in the case where the gap height (h) is 1.25 cm (resulting in a wider strand) is more prominent. However, the edges of the strand, in this case, are more aligned with the printing direction compared to the case where h is 1.75 cm.

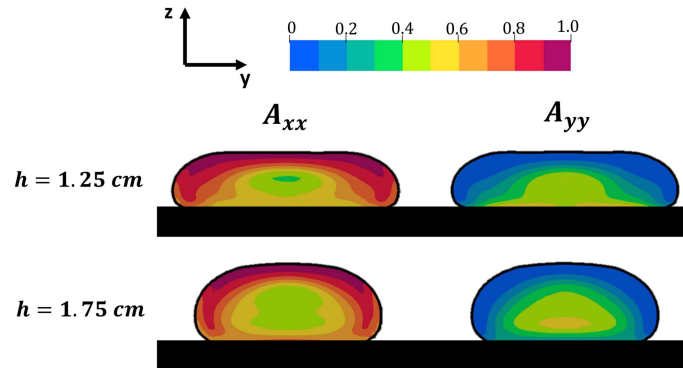


Figure 2: Fiber orientation at the end of the nozzle and during strand deposition.

This observation may seem unexpected, but it can be rationalized by considering the strand formation process. At the top of the strand, where the shearing effect from the nozzle is not yet present, there is a greater orientation of fibers in the printing direction. Furthermore, the vertical fibers (component A_{zz}) penetrate deeper into the material during the formation of the strand with h of 1.75 cm, resulting in a higher proportion of vertically oriented fibers and fewer fibers oriented in the printing direction. This unexpected behavior leads to a less oriented fiber orientation in the printing direction for strands with a larger gap height, which is intriguing considering that the lateral flow is less intense. The A_{xx} values for the case with h of 1.25 cm were 0.69, while for the case with h of 1.75 cm, they were 0.67. Although the difference is not significant, it was anticipated that with a more rounded shape, the fibers would align more in the printing direction, which was not observed.

In Fig. 2, it is evident that the fiber core region is larger in the case with a higher gap height, and the zones with highly oriented fibers are slightly smaller. This is primarily due to the fact that deformation mainly occurs in the central core region, with less impact on the sides of the strands. As the edges are already highly oriented in the printing direction, they occupy a larger spatial area in wider strands, resulting in slightly higher average fiber orientation in the printing direction for strands with h of 1.25 cm. Additionally, as shown in Fig. 3, the tall strands retain a greater presence of fibers oriented in the vertical direction, particularly in the central top region and at the very edges of the strands.

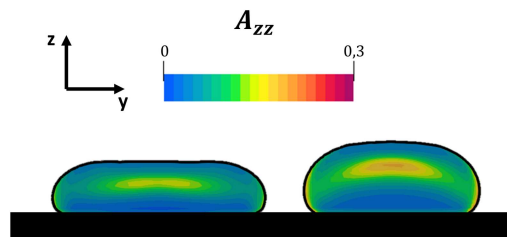


Figure 3: Fiber orientation at the end of the nozzle and during strand deposition.

3.2. Effect of multi-layer printing

When printing on a semi-solid surface, it is anticipated that the layers will experience deformation, particularly those situated at the bottom. This deformation occurs due to both the hydrostatic pressure generated by the weight of the freshly printed strands and the extrusion pressure in the nozzle during material extrusion. Having this in mind, it is crucial to comprehend the impact of this on fiber orientation and the degree to which deformation is connected to fiber reorientation. For this reason, we performed the simulations of two same cases from above, with different h values, up to two printed strands. The printed strands can be seen in Fig. 4.

Based on the simulations conducted, as shown in Table 1, the reorientation of fibers from a single-layer (SL) to a double-layer (DL) configuration was only observed in the case with a smaller h value. This can be attributed to the higher extrusion pressure exerted during printing with smaller gaps, leading to significant deformation of the first layer (see Fig. 4). The value of A_{xx} decreased from 0.69 to 0.66, primarily due to the changes in the bottom layer. Conversely, in the case where $h = 1.75$ cm, fiber orientation remained unchanged. While the bottom strand exhibited a slightly larger core region in the center, the newly deposited top strand displayed even greater alignment in the printing direction compared to a single-layer structure. When these two effects are taken into account in the calculation of average fiber orientation, it remains the same.

Table 1: Averaged fiber orientation in the cross.section

	$SL, h = 12.5\text{cm}$	$SL, h = 17.5\text{cm}$	$DL, h = 12.5\text{cm}$	$DL, h = 17.5\text{cm}$
A_{xx}	0.69	0.67	0.66	0.67
A_{yy}	0.26	0.25	0.29	0.25
A_{zz}	0.05	0.08	0.05	0.08

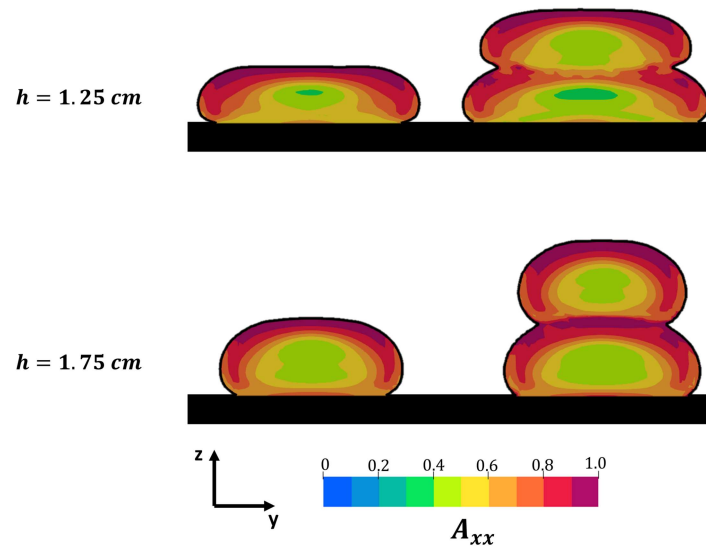


Figure 4: Fiber orientation at the end of the nozzle and during strand deposition.

3.3. Corner printing

Predicting fiber orientation during corner printing is a crucial task, considering that the optimal toolpath may be irregular since most complex shapes do not follow a straight line (26; 27). With this in mind, we are presenting numerical results for the first time, specifically focusing on corner printing at a 90 ° angle. As depicted in Fig. 5, the fibers at the cross-section of the corner exhibit an asymmetric orientation distribution. The fibers located at the outer corner are predominantly aligned in the direction from which the nozzle came, while those at the inner corner are aligned towards the direction in which the nozzle moves away.

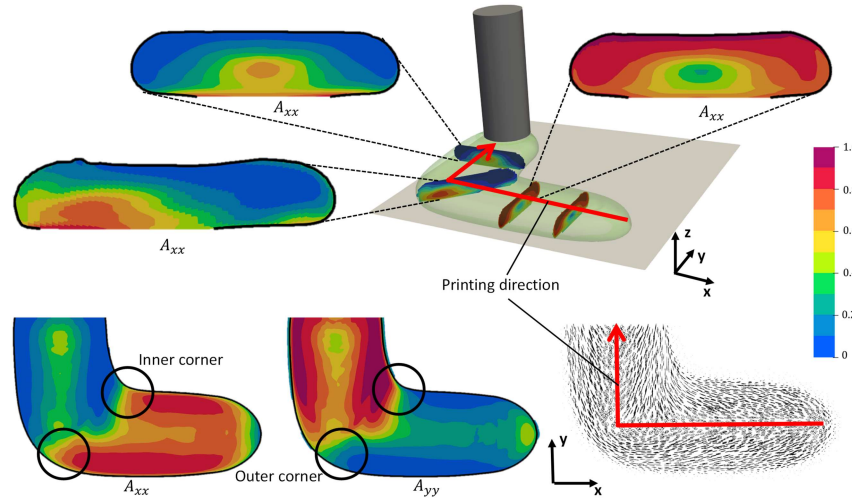


Figure 5: Fiber orientation during corner printing.

The observed effect is clearly visible in the cross-section displayed in Figure 6, which is captured at the central point of the corner. At this position, the cross-section appears significantly wider, indicating the occurrence of double deposition in the corner (28). While most of the material is compressed towards the inner corner, the fibers in that region tend to align themselves in the direction of the nozzle's movement after the corner print. This behavior can be attributed to the double shearing of the fibers by the nozzle exit, followed by the final shearing caused by the nozzle's movement in the y-direction. Consequently, the fibers orient themselves in the same direction as the nozzle's movement. Due to the over-extrusion at the inner corner, the fiber orientation at the corner becomes dependent on the printing direction. This leads to an interesting situation where, despite printing the same shape, the fiber orientation differs depending on the starting position. Understanding this concept makes it clear that strategically planning deceleration and acceleration at the corners to achieve more uniform deposition will also have a significant impact on fiber orientation at those corners.

4. Conclusions

This study has focused on investigating the orientation of fibers during various situations encountered in 3D printing. By analyzing different processing parameters and their influence on strand cross-section shape, the study has highlighted the correlation between printing conditions and fiber orientation. Additionally, the study has explored the impact of multiple layers on strand deformation and subsequent fiber reorientation. Notably, the study has examined the role of fiber orientation during corner printing and its potential for optimizing toolpath design.

The results have demonstrated that printing conditions play a significant role in determining fiber orientation. The ratio of extrusion/nozzle speed, in particular, has been shown to influence the shape of the strand and the lateral flow of the material, thereby affecting fiber orientation. Moreover, the study has emphasized that printed structures composed of multiple layers undergo deformation, leading to fiber reorientation. This observation underscores the importance of considering the effects of layer deposition on fiber orientation in 3D printing.

Furthermore, the study has presented numerical results for the first time regarding corner printing at a 90° angle. It has been observed that the fiber orientation in the cross-section of the corner is asymmetric, with fibers aligning differently at the outer and inner corners. This behavior is attributed to double shearing of the fibers by the nozzle exit and subsequent nozzle movement.

The findings of this study contribute to a better understanding of fiber orientation during 3D printing and highlight the importance of printing conditions and corner printing strategies in controlling and optimizing fiber orientation. Future research in this area could further explore the optimization of toolpaths to achieve more uniform deposition and investigate the mechanical properties of 3D printed structures considering fiber orientation as a key factor.

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References

- [1] M. Bayat, O. Zinovieva, F. Ferrari, C. Ayas, M. Langelaar, J. Spangenberg, R. Salajeghe, K. Poulis, S. Mohanty, O. Sigmund, et al., Holistic computational design within additive manufacturing through topology optimization combined with multiphysics multi-scale materials and process modelling, *Progress in Materials Science* (2023) 101129.
- [2] N. Su, R. S. Pierce, C. Rudd, X. Liu, Comprehensive investigation of reclaimed carbon fibre reinforced polyamide (rcf/pa) filaments and fdm printed composites, *Composites Part B: Engineering* 233 (2022) 109646.
- [3] M. Volk, O. Yuksel, I. Baran, J. H. Hattel, J. Spangenberg, M. Sandberg, Cost-efficient, automated, and sustainable composite profile manufacture: A review of the state of the art, innovations, and future of pultrusion technologies, *Composites Part B: Engineering* (2022) 110135.
- [4] H. L. Tekinalp, V. Kunc, G. M. Velez-Garcia, C. E. Duty, L. J. Love, A. K. Naskar, C. A. Blue, S. Ozcan, Highly oriented carbon fiber-polymer composites via additive manufacturing, *Composites Science and Technology* 105 (2014) 144–150.
- [5] R. Comminal, M. P. Serdeczny, D. B. Pedersen, J. Spangenberg, Numerical modeling of the strand deposition flow in extrusion-based additive manufacturing, *Additive Manufacturing* 20 (2018) 68–76.
- [6] M. T. Mollah, R. Comminal, M. P. Serdeczny, D. B. Pedersen, J. Spangenberg, Stability and deformations of deposited layers in material extrusion additive manufacturing, *Additive Manufacturing* 46 (2021) 102193.

- [7] M. T. Mollah, R. Comminal, M. P. Serdeczny, B. Šeta, J. Spangenberg, Computational analysis of yield stress buildup and stability of deposited layers in material extrusion additive manufacturing, *Additive Manufacturing* 71 (2023) 103605.
- [8] Z. Wang, D. E. Smith, A fully coupled simulation of planar deposition flow and fiber orientation in polymer composites additive manufacturing, *Materials* 14 (10) (2021) 2596.
- [9] E. Bertevas, J. Férec, B. C. Khoo, G. Ausias, N. Phan-Thien, Smoothed particle hydrodynamics (sph) modeling of fiber orientation in a 3d printing process, *Physics of Fluids* 30 (10) (2018) 103103.
- [10] V. Kumar, S. P. Alwekar, V. Kunc, E. Cakmak, V. Kishore, T. Smith, J. Lindahl, U. Vaidya, C. Blue, M. Theodore, et al., High-performance molded composites using additively manufactured preforms with controlled fiber and pore morphology, *Additive Manufacturing* 37 (2021) 101733.
- [11] B. Šeta, M. Sandberg, M. Brander, M. T. Mollah, D. Pokalla, V. Kumar, J. Spangenberg, Modelling fiber orientation during additive manufacturing-compression molding processes, *SFF symposium 2022* (2022).
- [12] B. Šeta, M. Brander, M. T. Mollah, V. NLN, D. Pokkalla, P. Kim, J. Spangenberg, Modeling of additive manufacturing-compression molding process using computational fluid dynamics, in: *Conference: 2022 Summer Topical Meeting: Advancing Precision in Additive Manufacturing - Knoxville, Tennessee, United States of America, 2022*.
- [13] B. Šeta, M. Sandberg, M. Brander, M. T. Mollah, D. Pokalla, V. Kumar, J. Spangenberg, Modeling fiber orientation and strand shape morphology in three-dimensional material extrusion additive manufacturing (Submitted).
- [14] J. Reinold, V. Gudžulić, G. Meschke, Computational modeling of fiber orientation during 3d-concrete-printing, *Computational Mechanics* (2023) 1–21.
- [15] P. Pibulchinda, E. Barocio, A. J. Favaloro, R. B. Pipes, Influence of printing conditions on the extrudate shape and fiber orientation in extrusion deposition additive manufacturing, *Composites Part B: Engineering* (2023) 110793.
- [16] B. Šeta, M. Sandberg, M. Brander, M. T. Mollah, D. Pokalla, V. Kumar, J. Spangenberg, Numerical modeling of fiber orientation in multi-layer, material-extrusion big area additive manufacturing (Submitted).
- [17] J. Yan, E. Demirci, A. Ganesan, A. Gleadall, Extrusion width critically affects fibre orientation in short fibre reinforced material extrusion additive manufacturing, *Additive Manufacturing* 49 (2022) 102496.
- [18] J. Yan, E. Demirci, A. Gleadall, 3d short fibre orientation for universal structures and geometries in material extrusion additive manufacturing, *Additive Manufacturing* 69 (2023) 103535.
- [19] E. Berberovic, Investigation of free-surface flow associated with drop impact: numerical simulations and theoretical modeling, Ph.D. thesis, Technische Universität (2010).
- [20] R. Comminal, W. R. L. da Silva, T. J. Andersen, H. Stang, J. Spangenberg, Modelling of 3d concrete printing based on computational fluid dynamics, *Cement and Concrete Research* 138 (2020) 106256.
- [21] N. Ranjbar, M. Mehrali, C. Kuenzel, C. Gundlach, D. B. Pedersen, A. Dolatshahi-Pirouz, J. Spangenberg, Rheological characterization of 3d printable geopolymers, *Cement and Concrete Research* 147 (2021) 106498.
- [22] M. T. Mollah, R. Comminal, W. R. Leal da Silva, B. Šeta, J. Spangenberg, Computational fluid dynamics modelling and experimental analysis of reinforcement bar integration in 3d concrete printing (Accepted in *Cement and Concrete Research*, 2023).

- [23] J. T. Pedersen-Bjergaard, M. M. Christensen, M. Brander, B. Šeta, D. B. Pedersen, J. Spangenberg, Interlayer mechanical properties of thermoset components produced by material extrusion additive manufacturing, SFF symposium 2022 (2022).
- [24] J. Férec, E. Bertevas, B. C. Khoo, G. Ausias, N. Phan-Thien, A rheological constitutive model for semiconcentrated rod suspensions in bingham fluids, *Physics of Fluids* 29 (7) (2017) 073103.
- [25] M. Jabbari, J. Spangenberg, J. H. Hattel, Particle migration using local variation of the viscosity (lvov) model in flow of a non-newtonian fluid for ceramic tape casting, *Chemical Engineering Research and Design* 109 (2016) 226–233.
- [26] F. Fernandez, W. S. Compel, J. P. Lewicki, D. A. Tortorelli, Optimal design of fiber reinforced composite structures and their direct ink write fabrication, *Computer Methods in Applied Mechanics and Engineering* 353 (2019) 277–307.
- [27] F. Fernandez, J. P. Lewicki, D. A. Tortorelli, Optimal toolpath design of additive manufactured composite cylindrical structures, *Computer Methods in Applied Mechanics and Engineering* 376 (2021) 113673.
- [28] M. T. Mollah, A. Moetazedian, A. Gleadall, J. Yan, W. E. Alphonso, R. Comminal, B. Šeta, T. Lock, J. Spangenberg, Investigation on corner precision at different corner angles in material extrusion additive manufacturing: An experimental and computational fluid dynamics analysis, in: 2022 International Solid Freeform Fabrication Symposium, 2022.